Unit

20 Oxidation and reduction

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Definitions of oxidation and reduction
- Common oxidizing and reducing agents
- Oxidation number
- Balanced redox equations

20.1 Defining oxidation and reduction in terms of gain and loss of oxygen

When a substance is burnt, it reacts with oxygen rapidly. Heat and light are also released during the process. For example, when charcoal (which is nearly pure carbon) is burnt, it reacts with oxygen to produce carbon dioxide.

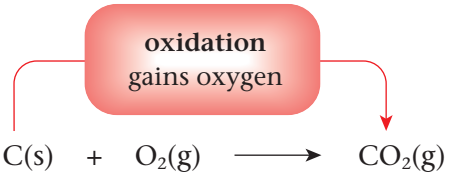


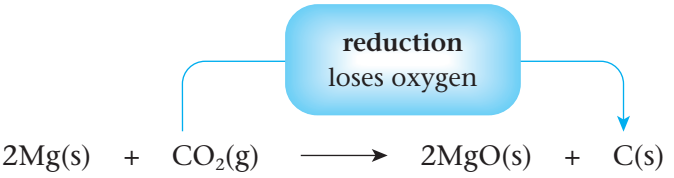


Fig. 20.1 Burning of charcoal is an oxidation process

The heat released is used to operate industrial processes, to warm homes, and for many other purposes.

Oxidation is a process in which a species gains oxygen. The burning of charcoal is an oxidation process (Fig. 20.1).

Magnesium burns in carbon dioxide to produce magnesium oxide and carbon according to the following equation:



Reduction is a process in which a species loses oxygen. Carbon dioxide undergoes a reduction in this case.

Look carefully at the above equation again. The magnesium gains oxygen to form magnesium oxide. It undergoes oxidation.

oxidation 氧化作用　　reduction 還原作用

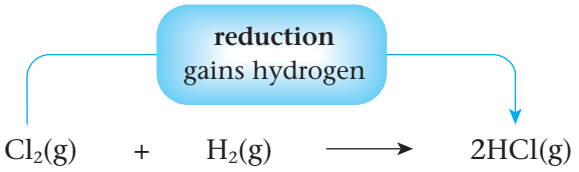
Reduction and oxidation always take place together. The combined process is called a **redox reaction** (**RED**uction + **OX**idation).

There are two other ways to define oxidation and reduction:

- in terms of the loss and gain of hydrogen; and
- in terms of the loss and gain of electrons.

20.2 Defining oxidation and reduction in terms of loss and gain of hydrogen

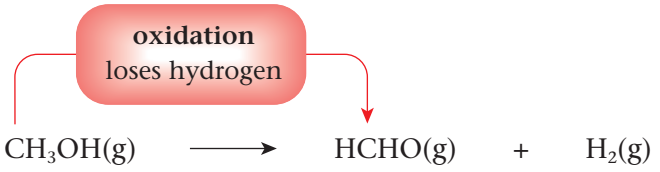
Chlorine reacts with hydrogen to form hydrogen chloride according to the following equation:



Reduction is a process in which a species gains hydrogen. Chlorine undergoes a reduction in this case.

When methanol (CH_3OH) is passed over a hot catalyst, methanal (HCHO) and hydrogen are produced according to the following equation:

Catalysts can increase the rates of reactions. We will discuss this in Topic 10 Rate of Reaction.



Oxidation is a process in which a species loses hydrogen. Methanol undergoes an oxidation in this case.

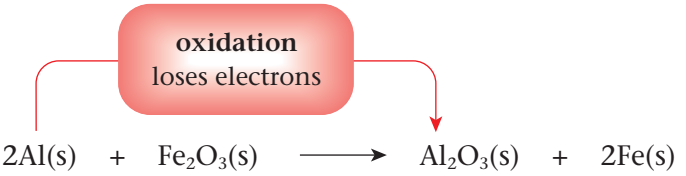
redox reaction 氧化還原反應



Fig. 20.2 The reaction between aluminium and iron(III) oxide is a very vigorous redox reaction

20.3 Defining oxidation and reduction in terms of loss and gain of electrons

Consider the reaction between aluminium and iron(III) oxide (Fig. 20.2):



In the reaction, aluminium undergoes oxidation as it gains oxygen.

During the oxidation process, each aluminium atom loses three electrons to become an aluminium ion.

$$Al \longrightarrow Al^{3+} + 3e^-$$

This gives us another definition for oxidation.

✓ Oxidation is a process in which a species loses electrons.

In the above reaction, iron(III) oxide undergoes reduction as it loses oxygen.

During the reduction process, each iron(III) ion gains three electrons to become an iron atom.

$$Fe^{3+} + 3e^- \longrightarrow Fe$$

✓ Reduction is a process in which a species gains electrons.

Do you know

Redox mnemonic

A mnemonic is a word or group of words used to help remember some information.

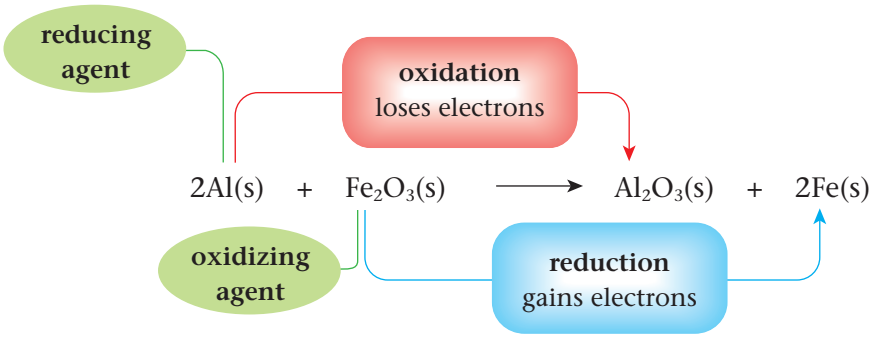
Let me introduce two friends to you:

LEO
(**L**ose **E**lectrons — **O**xidation);

GER
(**G**ain **E**lectrons — **R**eduction).

20.4 Oxidizing agent and reducing agent

In the redox reaction between aluminium and iron(III) oxide, aluminium causes the reduction of iron(III) oxide. Aluminium is the reducing agent. It reduces the other reactant and is itself oxidized at the same time.



On the other hand, iron(III) oxide causes the oxidation of aluminium. It is the oxidizing agent. It oxidizes the other reactant and is itself reduced at the same time.

✓ A species that causes reduction is a **reducing agent**.

✓ A species that causes oxidation is an **oxidizing agent**.

The characteristics of reducing and oxidizing agents are summarized in Table 20.1.

Table 20.1

Characteristics of reducing and oxidizing agents

Reducing agent	Oxidizing agent
causes reduction	causes oxidation
loses electrons	gains electrons
is oxidized	is reduced

reducing agent 還原劑　　oxidizing agent 氧化劑

20.5 Relative strength of reducing and oxidizing agents

We have looked at the electrochemical series of metals in Unit 19. Metals at the top of the series lose electrons easily. These metals are strong reducing agents.

CE 2006 Paper 2 Q29

Metals at the top of the electrochemical series (Table 20.2) form ions readily, which means their ions have little tendency to gain electrons. Thus, metal ions at the top of the electrochmeical series are weak oxidizing agents (Fig. 20.3).

By convention, the ionic half-equations are written from left to right as reductions, with the electrons on the left.

Hydrogen is not a metal.

Table 20.2

Trends of reducing power of metals and oxidizing power of metal ions

	ionic half-equation reduction →			
	Metal ion		Metal	
weak oxidizing agents	$K^+(aq) + e^-$	$\rightleftharpoons$	$K(s)$	strong reducing agents
	$Ca^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Ca(s)$	
	$Na^+(aq) + e^-$	$\rightleftharpoons$	$Na(s)$	
	$Mg^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Mg(s)$	
	$Al^{3+}(aq) + 3e^-$	$\rightleftharpoons$	$Al(s)$	
oxidizing power of metal ion increasing	$Zn^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Zn(s)$	reducing power of metal increasing
	$Fe^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Fe(s)$	
	$Pb^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Pb(s)$	
	$2H^+(aq) + 2e^-$	$\rightleftharpoons$	$H_2(g)$	
	$Cu^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Cu(s)$	
strong oxidizing agents	$Ag^+(aq) + e^-$	$\rightleftharpoons$	$Ag(s)$	weak reducing agents
	$Au^+(aq) + e^-$	$\rightleftharpoons$	$Au(s)$	



Fig. 20.3a Sodium ion in sodium chloride solution has little tendency to gain electrons and is a weak oxidizing agent



Fig. 20.3b Gold has little tendency to lose electrons and is a weak reducing agent

Practice 20.1

P20.1 Refer to Teacher's Resources Book page 40.

1 For each of the following conversions, decide whether an oxidation or a reduction is involved.

a) $Zn(s) \longrightarrow Zn^{2+}(aq)$

b) $Fe^{3+}(aq) \longrightarrow Fe^{2+}(aq)$

c) $2I^-(aq) \longrightarrow I_2(aq)$

2 Consider the following displacement reaction:

$Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

Complete the information below for each species.

	Cu(s)	$Ag^+(aq)$
Ionic half-equation for chemical change that occurs for the species		
Whether the species undergoes oxidation or reduction?		

20.6 Oxidation numbers

Consider the following reaction:

$$P_4(s) + 10Cl_2(g) \longrightarrow 4PCl_5(s)$$

This reaction does not involve oxygen and hydrogen. There is also no electron transfer. Is this a redox reaction? We can decide whether this is a redox reaction based on the concept of **oxidation number**◀.

Oxidation number is also known as oxidation state.

The oxidation number of an element is an imaginary charge assigned to it according to a set of rules. In assigning an oxidation number to each element in a compound, it is assumed that all compounds are ionic. N1

N1 Examination questions often ask students to give the oxidation numbers of certain elements in unfamiliar substances, e.g. the oxidation number of V in NH_4VO_3 (answer +5); the oxidation number of Pb in $[Pb(OH)_4]^{2-}$ (answer +2); the oxidation number of Co in $Co(NH_3)_4Cl_2$ (answer +2).

oxidation number 氧化數

Rules for assigning oxidation numbers

Table 20.3 provides a set of rules for assigning oxidation numbers.

State oxidation numbers in the format of +1, +2, etc,
✔ ✔
NOT 1+, 2+.
✘ ✘

The oxidation number of oxygen is –1 in peroxides, such as hydrogen peroxide (H_2O_2).

The oxidation number of hydrogen is –1 in metallic hydrides, such as sodium hydride (NaH).

ASL 2012 Paper 1 Q3(b)
AL 2012 Paper 1 Q3(b)(ii)

Table 20.3

Rules for assigning oxidation numbers

The oxidation number of ...	Examples
an element is 0.	For N_2, F_2, S and all metals: 0.
an element in a monoatomic ion is the same as the charge.	For Cu^{2+}: +2. For Cl^-: –1.
fluorine is –1 in its compounds.	For F in HF and PF_3: –1.
oxygen is usually –2 in its compounds (except in peroxides).	For O in H_2O and MgO: –2.
hydrogen is +1 in its covalent compounds (except in metal hydrides).	For H in H_2O and HCl: +1.
an alkali metal is +1 in its compounds.	For Na in NaCl and Na_2O: +1.
an alkaline earth metal is +2 in its compounds.	For Mg in MgO and MgF_2: +2.

There are two additional rules for assigning oxidation numbers:

1 The sum of oxidation numbers of all elements in a neutral compound is zero.

2 The sum of oxidation numbers of all elements in an ion is equal to the charge on the ion.

Example 20.1

Q Find the oxidation number of the underlined element in each of the following species:

a) $\underline{C}O_2$

b) $Na_2\underline{S}O_4$ N2 Refer to Teacher's Resources Book page 18.

c) $\underline{N}H_4^+$ N3 Refer to Teacher's Resources Book page 19.

CE 2010 Paper 2 Q11

d) $\underline{N}O_3^-$

e) $[\underline{Pb}(OH)_4]^{2-}$

Continued on next page

A a) $\underline{C}O_2$

Oxidation number of O = –2

Suppose the oxidation number of C in CO_2 is x.

$x + (-2) \times 2 = 0$

$x = +4$

∴ the oxidation number of C in CO_2 is +4.

b) $Na_2\underline{S}O_4$

Na_2SO_4 consists of Na^+ ions and SO_4^{2-} ion.

Suppose the oxidation number of S in SO_4^{2-} ion is x.

$x + (-2) \times 4 = -2$

$x = +6$

∴ the oxidation number of S in Na_2SO_4 is +6.

c) $\underline{N}H_4^+$

Suppose the oxidation number of N in NH_4^+ ion is x.

$x + (+1) \times 4 = +1$

$x = -3$

∴ the oxidation number of N in NH_4^+ ion is –3.

d) $\underline{N}O_3^-$

Suppose the oxidation number of N in NO_3^- ion is x.

$x + (-2) \times 3 = -1$

$x = +5$

∴ the oxidation number of N in NO_3^- ion is +5.

e) $[\underline{Pb}(OH)_4]^{2-}$

Suppose the oxidation number of Pb in $[Pb(OH)_4]^{2-}$ is x.

$x + (-2) \times 4 + (+1) \times 4 = -2$

$x = +2$

∴ the oxidation number of Pb in $[Pb(OH)_4]^{2-}$ is +2.

Some elements have different oxidation numbers in different species. For example, the oxidation numbers of nitrogen are –3 in NH_4^+ ion and +5 in NO_3^- ion.

Practice 20.2

P20.2 Refer to Teacher's Resources Book page 40.

Find the oxidation number of the underlined element in each of the following species.

a) $\underline{N}O$

b) $\underline{N}H_3$

c) $Na_2\underline{C}O_3$

d) $\underline{Cu}(OH)_2$

e) $\underline{Fe}^{3+}$

f) $\underline{Cr}_2O_7^{2-}$

20.7 Defining oxidation and reduction in terms of changes in oxidation numbers

Consider the following ionic half-equations that represent oxidation processes:

$$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$$

$$2Cl^-(aq) \longrightarrow Cl_2(aq) + 2e^-$$

Notice that the oxidation number of each species increases. The oxidation number of magnesium increases from 0 to +2 while that of chlorine increases from –1 to 0. This leads to another definition of oxidation.

✓ Oxidation is a process in which the oxidation number of an element in a species increases.

Consider the following ionic half-equations that represent reduction processes:

$$Ag^+(aq) + e^- \longrightarrow Ag(s)$$

$$Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq)$$

Notice that the oxidation number of each species decreases. The oxidation number of silver decreases from +1 to 0 while that of bromine decreases from 0 to –1.

✓ Reduction is a process in which the oxidation number of an element in a species decreases. N4

Do you know

Redox reactions in living organisms

The ability of carbon to take on different oxidation numbers is essential to life on Earth. Photosynthesis involves a series of reductions in which the oxidation number of carbon changes from +4 in carbon dioxide to 0 in sugars such as glucose. In cellular respiration, carbon undergoes a series of oxidations, after which the oxidation number of carbon is again +4 in carbon dioxide.

N4 Refer to Teacher's Resources Book page 19.

N5 Refer to Teacher's Resources Book page 20.

Using oxidation numbers to identify redox reactions N5

The oxidation number is a useful way of identifying whether a reaction is a redox reaction. If the oxidation numbers of the elements are changed during a reaction, it indicates that the reaction is a redox reaction (Fig. 20.4).

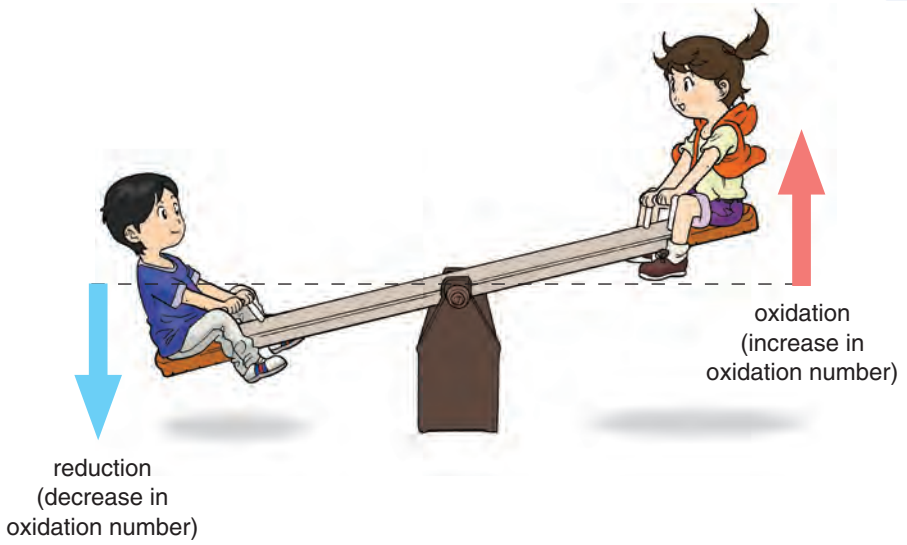


Fig. 20.4 Redox reactions involve changes in oxidation numbers

Example 20.2

Q For each of the following reactions, identify whether it is a redox reaction.

a) $PCl_3(l) + Cl_2(g) \longrightarrow PCl_5(s)$

b) $2I^-(aq) + H_2O_2(aq) + 2H^+(aq) \longrightarrow I_2(aq) + 2H_2O(l)$

c) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ — CE 2011 Paper 2 Q6

A a) The oxidation number of P increases from +3 to +5 while that of Cl decreases from 0 to –1. Thus, this is a redox reaction.

$\overset{+3}{PCl_3}(l) + \overset{0}{Cl_2}(g) \longrightarrow \overset{+5}{PCl_5}(s)$

b) The oxidation number of I increases from –1 to 0 while that of O decreases from –1 to –2. Thus, this is a redox reaction.

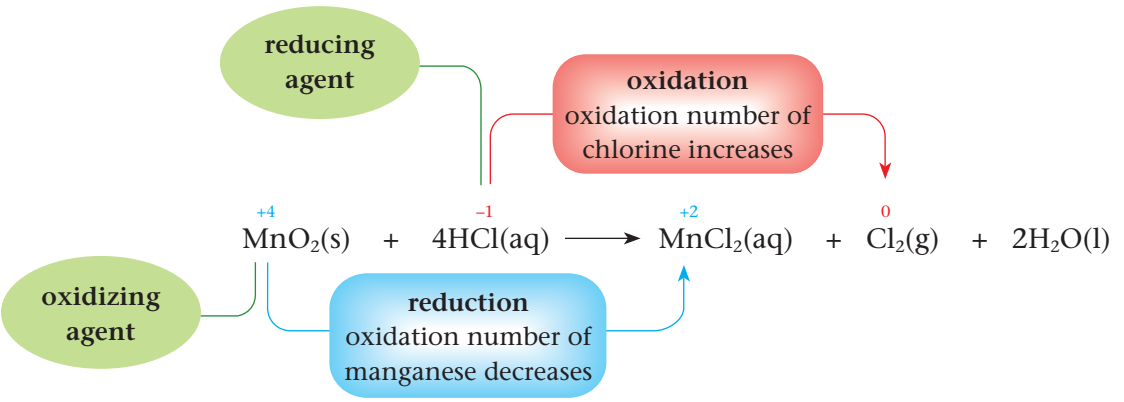
$\overset{-1}{2I^-}(aq) + \overset{-1}{H_2O_2}(aq) + 2H^+(aq) \longrightarrow \overset{0}{I_2}(aq) + \overset{-2}{2H_2O}(l)$

c) This is not a redox reaction because the oxidation numbers of all the elements remain unchanged in the reaction.

$\overset{+2}{Ca}\overset{+4}{C}\overset{-2}{O_3}(s) \longrightarrow \overset{+2}{Ca}\overset{-2}{O}(s) + \overset{+4}{C}\overset{-2}{O_2}(g)$

20.8 Using oxidation numbers to identify the oxidizing agent and reducing agent in a redox reaction

Manganese(IV) oxide reacts with concentrated hydrochloric acid according to the following equation:



The oxidation number of manganese decreases from +4 to +2 while that of chlorine increases from –1 to 0. Manganese(IV) oxide undergoes reduction in the reaction while concentrated hydrochloric acid undergoes oxidation.

- ✓ An oxidizing agent causes an increase in the oxidation number of an element in another species.
- ✓ A reducing agent causes a decrease in the oxidation number of an element in another species.

N6 Refer to Teacher's Resources Book page 21.

N6

Thus, manganese(IV) oxide is the oxidizing agent while concentrated hydrochloric acid is the reducing agent in the above reaction.

We can summarize the above ideas in Table 20.4.

Table 20.4

Using changes in oxidation numbers to identify the oxidizing agent and reducing agent

Oxidizing agent	Reducing agent
causes an increase in the oxidation number of an element in another species	causes a decrease in the oxidation number of an element in another species
is reduced	is oxidized
the oxidation number of a certain element in it decreases	the oxidation number of a certain element in it increases

Practice 20.3

P20.3 Refer to Teacher's Resources Book page 41.

1 For each of the following reactions, find out whether the oxidation number of the underlined element increases or decreases.

a) $2K\underline{N}O_3(s) \longrightarrow 2KNO_2(s) + O_2(g)$ CE 2010 Paper 2 Q24

b) $\underline{Cu}(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$

c) $\underline{Pb}O_2(s) + NO_2^-(aq) + 2H^+(aq) \longrightarrow Pb^{2+}(aq) + NO_3^-(aq) + H_2O(l)$ CE 2011 Paper 2 Q21

2 For each of the following reactions,

- determine whether it is a redox reaction;
- if the answer is yes, identify the oxidizing agent and reducing agent involved.

a) $2Al(s) + 6HCl(aq) \longrightarrow 2AlCl_3(aq) + 3H_2(g)$

b) $2CrO_4^{2-}(aq) + 2H^+(aq) \longrightarrow Cr_2O_7^{2-}(aq) + H_2O(l)$ CE 2010 Paper 2 Q24

c) $Mg(s) + CuSO_4(aq) \longrightarrow MgSO_4(aq) + Cu(s)$ CE 2011 Paper 2 Q6

20.9 Advantages and disadvantages of using the concept of oxidation number

Some reactions are not redox reactions, even though they involve ions. The concept of oxidation number helps us to decide whether a reaction is a redox reaction.

Consider the following reaction:

$$\overset{+1}{Ag^+}(aq) + \overset{-1}{Cl^-}(aq) \longrightarrow \overset{+1\ -1}{AgCl}(s)$$

The oxidation number of each element remains unchanged during the reaction. Hence this is not a redox reaction.

Besides, using oxidation numbers allows us to see exactly which part of a species is reduced or oxidized.

Consider the following equation:

$$\overset{-2}{5H_2S}(g) + 2\overset{+7}{KMnO_4}(aq) + 3H_2SO_4(aq) \longrightarrow \overset{0}{5S}(s) + 8H_2O(l) + 2\overset{+2}{MnSO_4}(aq) + K_2SO_4(aq)$$

It is obvious that sulphur is oxidized as its oxidation number increases from –2 to 0, and manganese is reduced because its oxidation number decreases from +7 to +2.

The main disadvantage of the concept of oxidation number is that it can cause a misunderstanding about the structure of molecular substances. The oxidation number of carbon in CO_2 is +4, but it does not mean that there is a charge of 4+ on the carbon atom.

Oxidation number is only an imaginary number.

20.10 The Stock system of naming compounds

The concept of oxidation number provides the basis for the *Stock system* of naming compounds.

Naming cations

Some metals, usually transition metals, have different oxidation numbers in different cations (Fig. 20.5). When naming such an ion, write a Roman numeral in brackets after the name of the metal to show its oxidation number. Table 20.5 lists some examples.

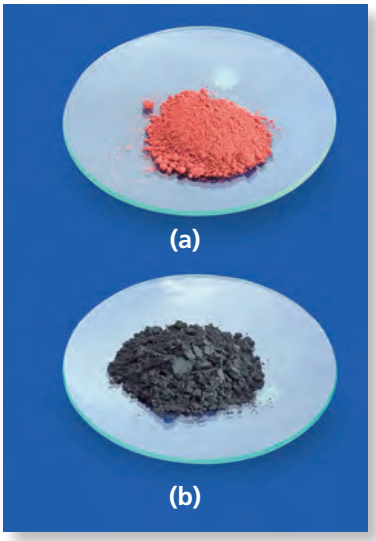


Fig. 20.5 (a) copper(I) oxide and (b) copper(II) oxide

Table 20.5

Some examples of naming cations	
Formula	**Stock name**
Cu_2O	copper(I) oxide
CuO	copper(II) oxide
$FeCl_2$	iron(II) chloride
$FeCl_3$	iron(III) chloride

Stock system 司托克系統

Naming polyatomic anions

Oxoanions are negative ions which contain oxygen. Under the Stock system, the name of an oxoanion often ends in -ate, followed by a Roman numeral in brackets to show the oxidation number of the central element. Table 20.6 lists some examples.

Under the Stock system, nitric acid (HNO_3) and nitrous acid (HNO_2) are named as nitric(V) acid and nitric(III) acid respectively. Sulphuric acid (H_2SO_4) and sulphurous acid (H_2SO_3) are named as sulphuric(VI) acid and sulphuric(IV) acid respectively.

Table 20.6

Some examples of naming oxoanions

Formula of species	Oxidation number of central element	Common name	Name by Stock system
MnO_4^-	+7	permanganate ion	manganate(VII) ion
MnO_4^{2-}	+6	manganate ion	manganate(VI) ion
$Cr_2O_7^{2-}$	+6	dichromate ion	dichromate(VI) ion
CrO_4^{2-}	+6	chromate ion	chromate(VI) ion
NO_3^-	+5	nitrate ion	nitrate(V) ion
NO_2^-	+3	nitrite ion	nitrate(III) ion
SO_4^{2-}	+6	sulphate ion	sulphate(VI) ion
SO_3^{2-}	+4	sulphite ion	sulphate(IV) ion

20.11 Common oxidizing and reducing agents

20.1
Investigating redox reactions.

Some common oxidizing and reducing agents and the products formed when they react are given in Tables 20.7 and 20.8. Figs. 20.6–20.8 show the colour changes of some oxidizing agents when they are reduced.

oxoanion 含氧陰離子

N7 When stating an observation with a colour change, it is necessary to give both the initial and final colours of the species concerned.



Fig. 20.6 $MnO_4^-(aq)$ ions (left) change to $Mn^{2+}(aq)$ ions (right) upon reduction



Fig. 20.7 $Cr_2O_7^{2-}(aq)$ ions (left) change to $Cr^{3+}(aq)$ ions (right) upon reduction

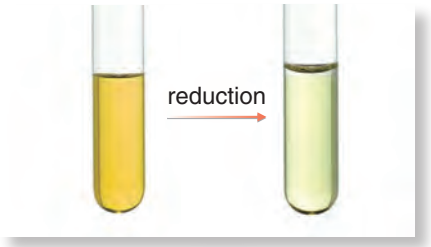


Fig. 20.8 $Fe^{3+}(aq)$ ions (left) change to $Fe^{2+}(aq)$ ions (right) upon reduction

Table 20.7

The products formed when some common oxidizing agents are reduced

Oxidizing agent	Product when reduced	Ionic half-equation N7
Metal ions low in the electrochemical series, e.g. $Ag^+(aq)$	$Ag(s)$	$Ag^+(g) + e^- \longrightarrow Ag(s)$ colourless → silvery
Chlorine	$Cl^-(aq)$	$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$ pale green → colourless
Aqueous bromine	$Br^-(aq)$	$Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq)$ yellow-brown → colourless
Potassium permanganate in acidic solution ◀	$Mn^{2+}(aq)$	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$ purple → pale pink (colourless)
Potassium dichromate in acidic solution ◀	$Cr^{3+}(aq)$	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$ orange → green
Iron(III) salts in solution, e.g. $Fe_2(SO_4)_3(aq)$	$Fe^{2+}(aq)$	$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$ yellow-brown → pale green
Concentrated nitric acid	$NO_2(g)$	$NO_3^-(aq) + 2H^+(aq) + e^- \longrightarrow NO_2(g) + H_2O(l)$ colourless → brown
Dilute nitric acid	$NO(g)$	$NO_3^-(aq) + 4H^+(aq) + 3e^- \longrightarrow NO(g) + 2H_2O(l)$ colourless → colourless
Concentrated sulphuric acid	$SO_2(g)$	$2H_2SO_4(l) + 2e^- \longrightarrow SO_4^{2-}(aq) + SO_2(g) + 2H_2O(l)$ colourless → colourless

◀ Potassium permanganate solution and potassium dichromate solution are usually acidified with dilute sulphuric acid.

Dilute nitric acid is NOT used because it is an oxidizing agent. It may react with the reducing agent in reactions.

Do you know

Breathalyzer – oxidation of ethanol by dichromate ions

A **breathalyzer** (Fig. 20.9) is a device for estimating alcohol content from a breath sample. To measure the level of alcohol in the breath, a suspect of drink-driving breathes into the device which contains a solution of dichromate ions. When the alcohol vapour makes contact with the solution, the colour of the solution changes from orange to green. The colour change is detected by a meter. The degree of the colour change is directly related to the level of alcohol in the suspect's breath.

Alcoholic drinks contain ethanol. During the reaction, ethanol (CH_3CH_2OH) is oxidized to ethanoic acid (CH_3COOH), while the orange dichromate ions are reduced to green chromium(III) ions.

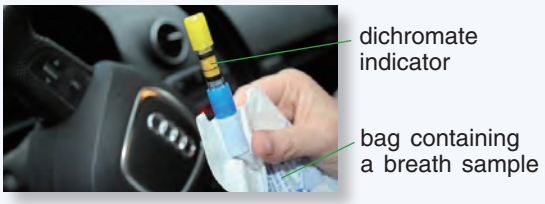


Fig. 20.9 A breathalyzer for estimating alcohol content from a breath sample

Oxidation: $CH_3CH_2OH(aq) + H_2O(l) \longrightarrow CH_3COOH(aq) + 4H^+(aq) + 4e^-$ CE 2011 Paper 1 Q10(b)(ii)

Reduction: $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

Table 20.8

The products formed when some common reducing agents are oxidized

Reducing agent	Product when oxidized	Ionic half-equation
Metals high in the electrochemical series, e.g. K(s)	$K^+(aq)$	$K(s) \longrightarrow K^+(aq) + e^-$ (silvery → colourless)
Carbon	$CO_2(g)$ $CO(g)$	Not applicable
Sulphite solutions	$SO_4^{2-}(aq)$	$SO_3^{2-}(aq) + H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$ (colourless → colourless)
Iron(II) salts in solution, e.g. $FeSO_4(aq)$	$Fe^{3+}(aq)$	$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$ N8 (pale green → yellow-brown)
Bromide solutions, e.g. KBr(aq)	$Br_2(aq)$	$2Br^-(aq) \longrightarrow Br_2(aq) + 2e^-$ (colourless → yellow-brown)
Iodide solutions, e.g. KI(aq)	$I_2(aq)$	$2I^-(aq) \longrightarrow I_2(aq) + 2e^-$ (colourless → brown)

N8 Refer to Teacher's Resources Book page 22.

breathalyzer 呼氣分析儀

Practice 20.4 P20.4 Refer to Teacher's Resources Book page 41.

Consider the reaction between acidified potassium dichromate solution and iron(II) sulphate solution.

Complete the information below for each reactant involved.

	Acidified potassium dichromate solution	Iron(II) sulphate solution
Product formed when it reacts		
Ionic half-equation		
As an oxidizing agent or a reducing agent?		

You need to study either Section 20.12 or 20.13, but not both.

20.12 Balancing redox equations using ionic half-equations

Many redox reactions can result from various combinations of oxidizing and reducing agents. We can obtain balanced equations for these redox reactions by combining ionic half-equations that describe the chemical changes of the oxidizing and reducing agents. The main rule is to balance the number of electrons lost in oxidation with that gained in reduction.

The rules are summarized below:

1 Write down the oxidizing agent and reducing agent involved. Determine their products.

2 Write separate ionic half-equations for the oxidation and reduction. Balance each ionic half-equation with respect to the number of atoms and charges.

a) In an acidic solution, balance the number of oxygen or hydrogen atoms by adding the proper number of H_2O or H^+ to either side of the ionic half-equation.

b) Then balance the charge of each ionic half-equation by adding electron(s) to either side of the equation.

3 Multiply each balanced ionic half-equation by a number so that the number of electrons gained in one ionic half-equation is equal to that lost in the other.

4 Combine the two ionic half-equations and eliminate the electrons. Collect like terms if necessary. This gives the balanced redox equation.

Example 20.3

Q Write a balanced redox equation for the reaction between acidified potassium permanganate solution and iron(II) sulphate solution.



acidified potassium permanganate solution | iron(II) sulphate solution | reaction mixture

A

1 Write down the oxidizing agent and the reducing agent involved.	Acidified $KMnO_4$ is the oxidizing agent and $FeSO_4$ is the reducing agent.
2 a) Write an ionic half-equation for the reduction process.	The unbalanced ionic half-equation for acidified $KMnO_4$ is: $MnO_4^-(aq) \longrightarrow Mn^{2+}(aq)$
i) Balance the ionic half-equation with respect to the number of atoms.	To balance the 4 oxygen atoms in MnO_4^-, add $4H_2O$ on the right-hand side. $MnO_4^-(aq) \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$ To balance the 8 hydrogen atoms in $4H_2O$, add $8H^+$ on the left-hand side. $MnO_4^-(aq) + 8H^+(aq) \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$
ii) Balance the ionic half-equation with respect to the number of charges.	Charge on left-hand side = (–1) + 8 x (+1) = +7 Charge on right-hand side = +2 ∴ add $5e^-$ on the left-hand side to balance the charge. The balanced ionic half-equation for acidified $KMnO_4$ is: $MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$.....(i)

Continued on next page

b) Write an ionic half-equation for the oxidation process.	The unbalanced ionic half-equation for $FeSO_4$ is: $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq)$
i) Balance the ionic half-equation with respect to the number of atoms.	The number of atoms is balanced.
ii) Balance the ionic half-equation with respect to the number of charges.	Charge on left-hand side = +2 Charge on right-hand side = +3 $\therefore$ add $1e^-$ on the right-hand side to balance the charge. The balanced ionic half-equation for $FeSO_4$ is: $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$ .....(ii)
3 Make the number of electrons gained in one ionic half-equation equal to that lost in the other.	Multiply equation (ii) by 5.
4 Combine the two ionic half-equations and eliminate the electrons.	(i) $MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$ (ii) x 5 $5Fe^{2+}(aq) \longrightarrow 5Fe^{3+}(aq) + 5e^-$ $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$

Example 20.4

Q Write a balanced redox equation for the reaction between acidified potassium dichromate solution and sodium sulphite solution.



acidified potassium dichromate solution | sodium sulphite solution | reaction mixture

Continued on next page

A1

1 Write down the oxidizing agent and the reducing agent involved.	Acidified $K_2Cr_2O_7$ is the oxidizing agent and Na_2SO_3 is the reducing agent.
2 a) Write an ionic half-equation for the reduction process.	The unbalanced ionic half-equation for acidified $K_2Cr_2O_7$ is: $Cr_2O_7^{2-}(aq) \longrightarrow 2Cr^{3+}(aq)$
i) Balance the ionic half-equation with respect to the number of atoms.	To balance the 7 oxygen atoms in $Cr_2O_7^{2-}$, add $7H_2O$ on the right-hand side. $Cr_2O_7^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$ To balance the 14 hydrogen atoms in $7H_2O$, add $14H^+$ on the left-hand side. $Cr_2O_7^{2-}(aq) + 14H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$
ii) Balance the ionic half-equation with respect to the number of charges.	Charge on left-hand side = (–2) + 14 x (+1) = +12 Charge on right-hand side = 2 x (+3) = +6 $\therefore$ add $6e^-$ on the left-hand side to balance the charge. The balanced ionic half-equation for acidified $K_2Cr_2O_7$ is: $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$.....(i)
b) Write an ionic half-equation for the oxidation process.	The unbalanced ionic half-equation for Na_2SO_3 is: $SO_3^{2-}(aq) \longrightarrow SO_4^{2-}(aq)$
i) Balance the ionic half-equation with respect to the number of atoms.	To balance the oxygen atoms, add H_2O on the left-hand side. $SO_3^{2-}(aq) + H_2O(l) \longrightarrow SO_4^{2-}(aq)$ To balance the 2 hydrogen atoms in H_2O, add $2H^+$ on the right-hand side. $SO_3^{2-}(aq) + H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq)$
ii) Balance the ionic half-equation with respect to the number of charges.	Charge on left-hand side = –2 Charge on right-hand side = (–2) + 2 x (+1) = 0 $\therefore$ add $2e^-$ on the right-hand side to balance the charge. The balanced ionic half-equation for Na_2SO_3 is: $SO_3^{2-}(aq) + H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$ .....(ii)
3 Make the number of electrons gained in one ionic half-equation equal to that lost in the other.	Multiply equation (ii) by 3.
4 Combine the two ionic half-equations and eliminate the electrons.	(i) $Cr_2O_7^{2-}(aq) + 14H^+(aq) + \cancel{6e^-} \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$ (ii) x 3 $3SO_3^{2-}(aq) + 3H_2O(l) \longrightarrow 3SO_4^{2-}(aq) + 6H^+(aq) + \cancel{6e^-}$ $Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 14H^+(aq) + 3H_2O(l) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + 7H_2O(l) + 6H^+(aq)$ Since $H^+(aq)$ and $H_2O(l)$ appear on both sides of the equation, simplify the equation by collecting like terms. $Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + 4H_2O(l)$

20.13 Balancing redox equations using oxidation number method

There is another method for balancing redox equations, known as the oxidation number method. The rules are as follows:

1 Write down the oxidizing agent and reducing agent involved. Determine their products.

2 Assign oxidation numbers to all atoms.

3 Notice those atoms which undergo a change in oxidation number. Determine the number of electrons lost or gained per formula unit.

4 Insert an appropriate coefficient before the formula of each reagent on the left-hand side of the equation to make the number of electrons gained by the oxidizing agent equal to that lost by the reducing agent.

5 Add appropriate coefficients on the right-hand side to balance the number of atoms which have gained or lost electrons.

6 Balance the numbers of all other atoms except O and H.

7 Add either H^+ or OH^- to make the number of charges (the sum of all ionic charges) on both sides of the equation equal. If the reaction takes place in acidic solution, add H^+ to the side *deficient* in positive charges.

8 Add H_2O to the appropriate side to balance the number of O atoms. Check to see that the number of H atoms is now balanced.

deficient 缺乏

Example 20.5

Q Write a balanced redox equation for the reaction between acidified potassium permanganate solution and iron(II) sulphate solution.

A

1 Write down the oxidizing agent and the reducing agent involved. Determine their products.	Acidified $KMnO_4$ is the oxidizing agent and $FeSO_4$ is the reducing agent. $MnO_4^-(aq) \longrightarrow Mn^{2+}(aq)$ $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq)$
2 Assign oxidation numbers to all atoms.	$\overset{+7}{MnO_4^-}(aq) + \overset{+2}{Fe^{2+}}(aq) \longrightarrow \overset{+2}{Mn^{2+}}(aq) + \overset{+3}{Fe^{3+}}(aq)$
3 Notice atoms which undergo a change in oxidation number. Determine the number of electrons lost or gained per formula unit.	reduction: gain of $5e^-$ per MnO_4^- $\overset{+7}{MnO_4^-}(aq) + \overset{+2}{Fe^{2+}}(aq) \longrightarrow \overset{+2}{Mn^{2+}}(aq) + \overset{+3}{Fe^{3+}}(aq)$ oxidation: loss of $1e^-$ per Fe^{2+}
4 Insert an appropriate coefficient before the formula of each reagent on the left-hand side to make the number of electrons gained equal to that lost.	$MnO_4^-(aq) + 5Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + Fe^{3+}(aq)$
5 Add appropriate coefficients on the right-hand side to balance the number of atoms which have gained or lost electrons.	$MnO_4^-(aq) + 5Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq)$
6 Balance the numbers of all other atoms except O and H.	The numbers of all other atoms except O are balanced.
7 Add H^+ to the side deficient in positive charges to make the number of charges on both sides equal.	Total charge on left-hand side = (–1) + 5 x (+2) = +9 Total charge on right-hand side = (+2) + 5 x (+3) = +17 Add $8H^+$ to the left-hand side. $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq)$
8 Add H_2O to the appropriate side to balance the number of O atoms.	$MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \longrightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$

Example 20.6

Q Write a balanced redox equation for the reaction between acidified potassium dichromate solution and sodium sulphite solution.

A

1 Write down the oxidizing agent and the reducing agent involved. Determine their products.	Acidified $K_2Cr_2O_7$ is the oxidizing agent and Na_2SO_3 is the reducing agent. $Cr_2O_7^{2-}(aq) \longrightarrow Cr^{3+}(aq)$ $SO_3^{2-}(aq) \longrightarrow SO_4^{2-}(aq)$
2 Assign oxidation numbers to all atoms.	$\overset{+6}{Cr_2O_7^{2-}}(aq) + \overset{+4}{SO_3^{2-}}(aq) \longrightarrow \overset{+3}{Cr^{3+}}(aq) + \overset{+6}{SO_4^{2-}}(aq)$
3 Notice atoms which undergo a change in oxidation number. Determine the number of electrons lost or gained per formula unit.	reduction: gain of $6e^-$ per $Cr_2O_7^{2-}$ $\overset{+6}{Cr_2O_7^{2-}}(aq) + \overset{+4}{SO_3^{2-}}(aq) \longrightarrow \overset{+3}{Cr^{3+}}(aq) + \overset{+6}{SO_4^{2-}}(aq)$ oxidation: loss of $2e^-$ per SO_3^{2-}
4 Insert an appropriate coefficient before the formula of each reagent on the left-hand side to make the number of electrons gained equal to that lost.	$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) \longrightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$
5 Add appropriate coefficients on the right-hand side to balance the number of atoms which have gained or lost electrons.	$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq)$
6 Balance the numbers of all other atoms except O and H.	The numbers of all other atoms except O are balanced.
7 Add H^+ to the side deficient in positive charges to make the number of charges on both sides equal.	Total charge on left-hand side = (–2) + 3 x (–2) = –8 Total charge on right-hand side = 2 x (+3) + 3 x (–2) = 0 Add $8H^+$ to the left-hand side. $Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq)$
8 Add H_2O to the appropriate side to balance the number of O atoms.	$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + 4H_2O(l)$

There is a gain of $3e^-$ per Cr.

Example 20.7

Q Hydrogen peroxide (H_2O_2) is an oxide of hydrogen. N9 Refer to Teacher's Resources Book page 23.

a) Hydrogen peroxide decomposes at room temperature according to the following equation:

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

i) What is the oxidation number of oxygen in hydrogen peroxide?

ii) What happens to the oxidation number of oxygen during the decomposition?

b) In the presence of a dilute acid, hydrogen peroxide reacts with iodide ions. Iodine and water are produced.

i) Write an ionic half-equation for the formation of water from hydrogen peroxide.

ii) Write an ionic half-equation for the chemical change of iodide ions.

iii) Write an ionic equation for the reaction between hydrogen peroxide and iodide ions.

iv) Decide whether hydrogen peroxide acts as an oxidizing agent or a reducing agent in the reaction. Explain your answer.

v) State the expected observation.

A a) i) –1

ii) The oxidation number of oxygen increases from –1 to 0, and decreases from –1 to –2 at the same time.

b) i) $H_2O_2(aq) + 2H^+(aq) + 2e^- \longrightarrow 2H_2O(l)$

ii) $2I^-(aq) \longrightarrow I_2(aq) + 2e^-$

iii) $H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \longrightarrow I_2(aq) + 2H_2O(l)$

iv) Hydrogen peroxide acts as an oxidizing agent. The oxidation number of oxygen decreases from –1 to –2.

v) The colour of the reaction mixture changes from colourless to brown.

Practice 20.5

P20.5 Refer to Teacher's Resources Book page 41.

1 Zinc can reduce dichromate ions in acidic solution to chromium(III) ions.

The relevant ionic half-equations are as follows:

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

$$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$$

Construct an ionic equation for the reaction.

2 Consider the reaction between acidified potassium permanganate solution and potassium sulphite solution.

a) Write ionic half-equations for the reduction and oxidation.

b) Write an ionic equation for the reaction.

c) Identify the oxidizing agent and the reducing agent.

d) State ONE observation.

20.14 The electrochemical series and the relative oxidizing / reducing power of common oxidizing / reducing agents

We have looked at the electrochemical series of metals earlier in this unit. Table 20.9 shows a version of the series that also includes common oxidizing and reducing agents.

Each ionic half-equation in the electrochemical series is written in this form:

oxidized form + ne^- $\rightleftharpoons$ reduced form

Table 20.9

A detailed version of the electrochemical series that includes common oxidizing and reducing agents

	ionic half-equation	
	reduction → / ← oxidation	
	Oxidized form	**Reduced form**
weak oxidizing agents	$Li^+(aq) + e^- \rightleftharpoons Li(s)$	strong reducing agents
	$K^+(aq) + e^- \rightleftharpoons K(s)$	
	$Ca^{2+}(aq) + 2e^- \rightleftharpoons Ca(s)$	
	$Na^+(aq) + e^- \rightleftharpoons Na(s)$	
	$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s)$	
	$Al^{3+}(aq) + 3e^- \rightleftharpoons Al(s)$	
	$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	
	$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	
	$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s)$	
oxidizing power increasing	$2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$	reducing power increasing
	$H_2SO_4(l) + 2H^+(aq) + 2e^- \rightleftharpoons SO_2(g) + 2H_2O(l)$	
	$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	
	$O_2(g) + 2H_2O(l) + 4e^- \rightleftharpoons 4OH^-(aq)$	
	$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$	
	$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	
	$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	
	$Br_2(aq) + 2e^- \rightleftharpoons 2Br^-(aq)$	
	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l)$	
	$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq)$	
strong oxidizing agents	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$	weak reducing agents
	$F_2(g) + 2e^- \rightleftharpoons 2F^-(aq)$	

Notice the following points about the electrochemical series:

- The oxidizing power of oxidizing agents increases down the series.
- The reducing power of reducing agents decreases down the series.
- The strongest oxidizing agent is $F_2(g)$ while the strongest reducing agent is Li(s).

As the oxidizing and reducing agents are arranged in order of their oxidizing and reducing powers in the series, we can use the series to predict possible redox reactions.

Predicting the feasibility of a redox reaction when chlorine gas is passed into iron(II) sulphate solution

To predict the feasilibity of a redox reaction when chlorine gas is passed into iron(II) sulphate solution (Fig. 20.10), look at the two relevant ionic half-equations in the electrochemical series:

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$$

$$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq)$$

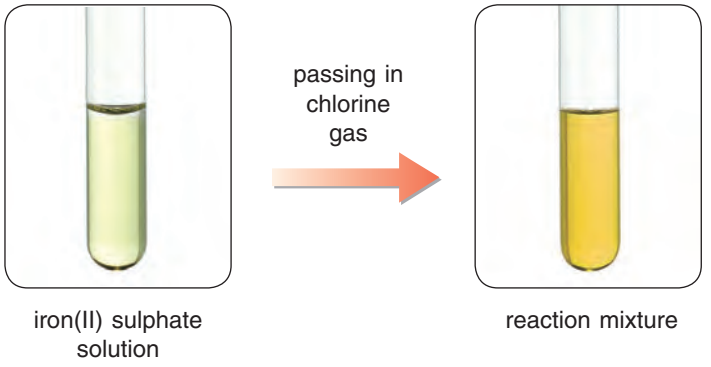


Fig. 20.10 Passing $Cl_2(g)$ into $FeSO_4(aq)$

Fig. 20.11 shows the relative positions of the two ionic half-equations in the electrochemical series. As $Cl_2(g)$ is a stronger oxidizing agent than $Fe^{3+}(aq)$ ion, it will oxidize the $Fe^{2+}(aq)$ ions to $Fe^{3+}(aq)$ ions.

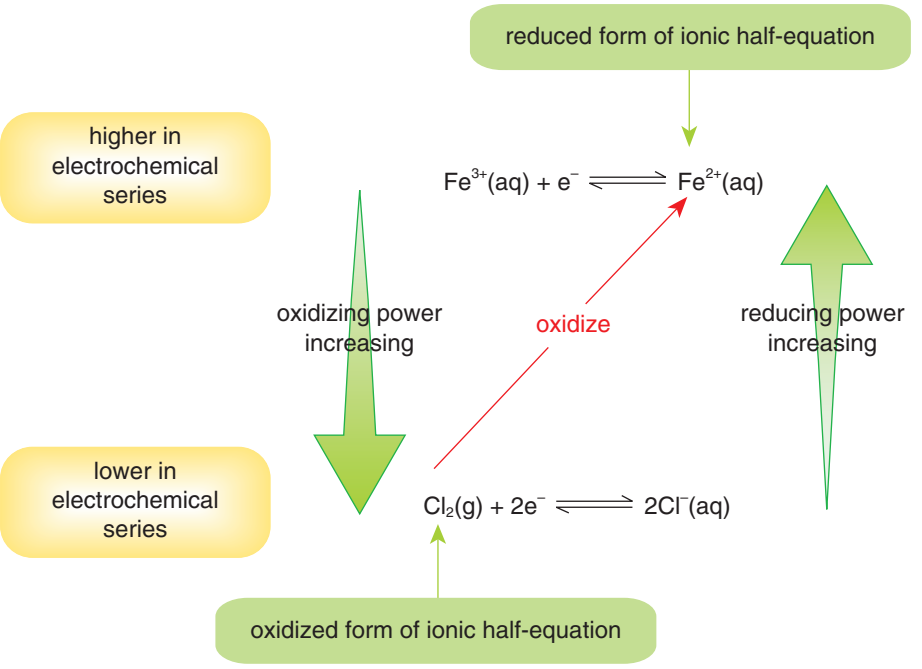


Fig. 20.11 The relative positions of the two ionic half-equations in the electrochemical series

Fig. 20.12 summarizes the possible reaction when chlorine gas is passed into iron(II) sulphate solution.

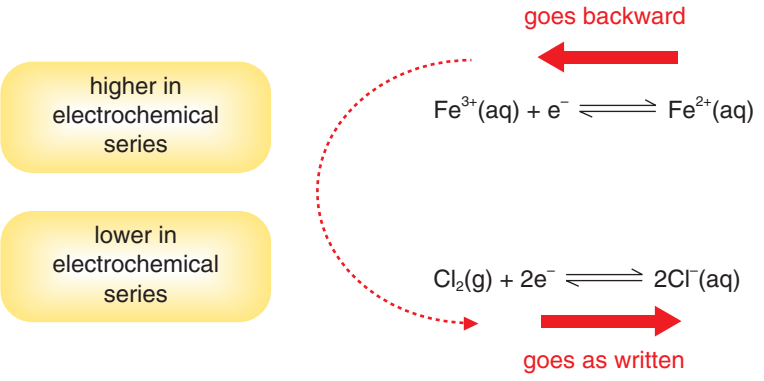


Fig. 20.12 Possible reaction when chlorine gas is passed into iron(II) sulphate solution

Combine the two ionic half-equations to obtain the ionic equation for the possible reaction:

$$Cl_2(g) + 2Fe^{2+}(aq) \longrightarrow 2Cl^-(aq) + 2Fe^{3+}(aq)$$

Example 20.8

Q The tendency of being reduced of three ionic species increases in the order as shown below:

$Na^+(aq) < H^+(aq) < Hg^{2+}(aq)$

Based on the above information, predict whether the following statement is correct. Explain your answer.

'Na(s) is a stronger reducing agent than Hg(l).'

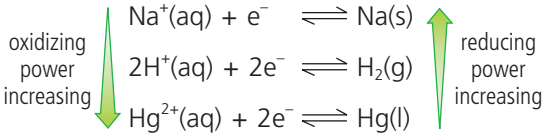
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A The statement is correct.

The $Hg^{2+}(aq)$ ion has the greatest tendency to be reduced while the $Na^+(aq)$ ion has the least tendency. It can be deduced that the $Hg^{2+}(aq)$ ion has the strongest oxidizing power while the $Na^+(aq)$ ion has the weakest oxidizing power.

Thus, Na(s) has a stronger reducing power than Hg(l).

Based on the given information, it is possible to deduce the relative positions of the ionic species in the electrochemical series.



20.15 Chlorine as an oxidizing agent N10

Chlorine is a powerful oxidizing agent. Chlorine atoms gain electrons readily to form chloride ions (Cl^-). The oxidation number of chlorine decreases from 0 to –1 in a redox reaction. N11

$$\overset{0}{Cl_2} + 2e^- \longrightarrow \overset{-1}{2Cl^-}$$

The oxidizing power of the halogens is in the order of $Cl_2 > Br_2 > I_2$.

N10 Examination questions often ask about aqueous bromine.
- Aqueous bromine is a common oxidizing agent.
- It can react with NaI(aq).
- It changes from yellow-brown to colourless when reduced.
  $Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq)$

N11 Aqueous chlorine can convert iron(II) ions to iron(III) ions.

chlorine 氯

20.2

Ranking halogens according to their oxidizing power.

Video

Ranking halogens according to their oxidizing power

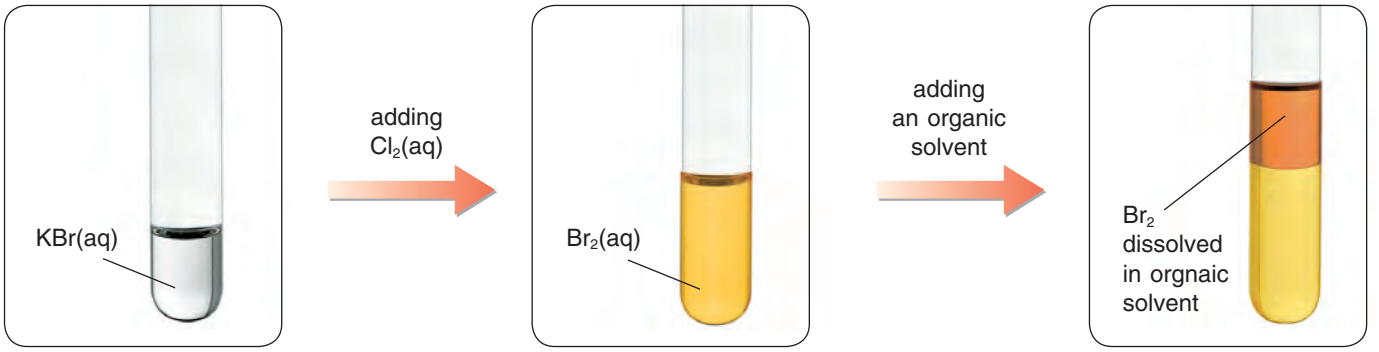
Video

Reaction of aqueous chlorine with halide solutions

Action of aqueous chlorine on potassium bromide solution N12

Chlorine dissolves in water to form a very pale green solution. When we add aqueous chlorine to potassium bromide solution, the solution becomes yellow-brown in colour (Fig. 20.13b). Bromine is formed in this reaction.

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(a) Colourless potassium bromide solution

(b) Mixture is yellow-brown in colour due to the presence of bromine

(c) Bromine gives an orange colour when dissolved in the organic solvent

Fig. 20.13 Reaction between aqueous chlorine and potassium bromide solution, followed by addition of an organic solvent

N12 Halide ions as reducing agents

The strength of halide salts as reducing agents increases from Cl^- to I^-.

The ionic half-equation for the reduction action of a halide ion X^- is:

$2X^-(aq) \longrightarrow X_2(aq) + 2e^-$

- Chloride ions are very difficult to oxidize. They are only oxidized by very strong oxidizing agents such as acidified potassium permanganate solution.
- Bromide ions are more easily oxidized. Chlorine, acidified potassium dichromate solution and potassium permanganate solution all oxidize bromide ions. .
- Iodide ions are strong reducing agents and are oxidized by many substances. For example, iron(III) ions are reduced to iron(II) ions by iodide ions in solution. The two ionic half-equations are:

  $2I^-(aq) \longrightarrow I_2(aq) + 2e^-$

  $Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$

  The overall equation is:

  $Fe^{3+}(aq) + 2I^-(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(aq)$

N13 Displacement reactions are redox reactions.

In this reaction, chlorine atoms gain electrons and bromide ions lose electrons.

$$Cl_2(aq) + 2e^- \longrightarrow 2Cl^-(aq)$$

$$2Br^-(aq) \longrightarrow Br_2(aq) + 2e^-$$

Chlorine is a stronger oxidizing agent than bromine. It oxidizes the bromide ions to bromine. N13

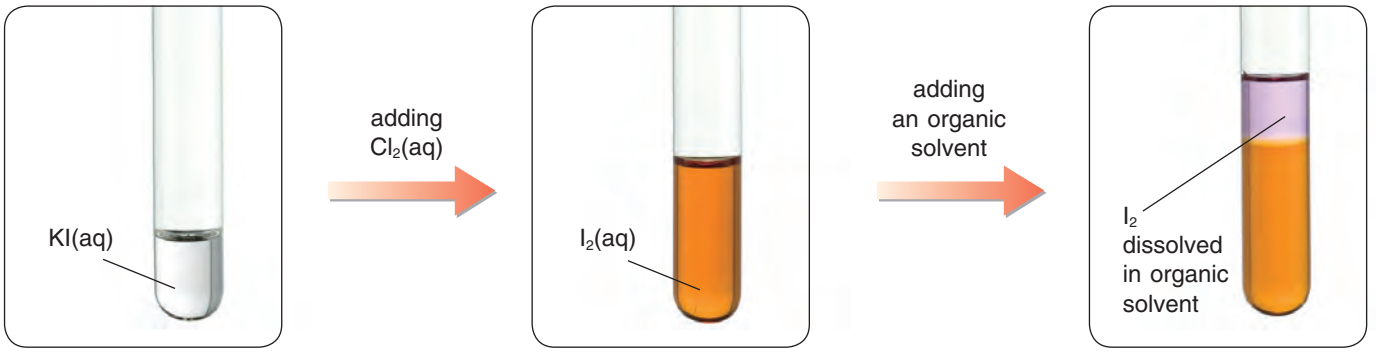
$$\underset{\text{very pale green}}{Cl_2(aq)} + \underset{\text{colourless}}{2Br^-(aq)} \longrightarrow \underset{\text{colourless}}{2Cl^-(aq)} + \underset{\text{yellow-brown}}{Br_2(aq)}$$

Bromine gives an orange colour when it is dissolved in an organic solvent. To confirm that the yellow-brown substance formed in the reaction is bromine, add an organic solvent to the reaction mixture. Bromine dissolves in the solvent and an orange layer appears (Fig. 20.13c).

Action of aqueous chlorine on potassium iodide solution

CE 2007 Paper 2 Q19
CE 2009 Paper 2 Q31

When we add aqueous chlorine to potassium iodide solution, the solution becomes brown in colour (Fig. 20.14b). Iodine is formed in the reaction.



(a) Colourless potassium iodide solution

(b) Mixture is brown in colour due to the presence of iodine

(c) Iodine gives a violet colour when dissolved in the organic solvent

Fig. 20.14 Reaction between aqueous chlorine and potassium iodide solution, followed by addition of an organic solvent

In this reaction, chlorine atoms gain electrons and iodide ions lose electrons.

$$Cl_2(aq) + 2e^- \longrightarrow 2Cl^-(aq)$$

$$2I^-(aq) \longrightarrow I_2(aq) + 2e^-$$

Chlorine is a stronger oxidizing agent than iodine. It oxidizes the iodide ions to iodine.

$$\underset{\text{very pale green}}{Cl_2(aq)} + \underset{\text{colourless}}{2I^-(aq)} \longrightarrow \underset{\text{colourless}}{2Cl^-(aq)} + \underset{\text{brown}}{I_2(aq)}$$

CE 2007 Paper 1 Q3(b)(i)

Iodine gives a violet colour when it is dissolved in an organic solvent. To confirm that the brown substance formed in the reaction is iodine, add an organic solvent to the reaction mixture. Iodine dissolves in the solvent and a violet layer appears (Fig. 20.14c).

Do you know

Household bleach

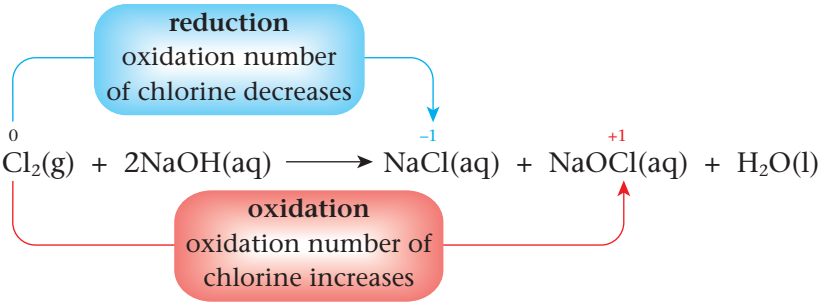
Household bleach is made by reacting chlorine with sodium hydroxide solution. The active ingredient in household bleach is sodium hypochlorite, NaOCl.

Household bleach should NOT be used together with acids because $OCl^-(aq)$ ions react with $H^+(aq)$ ions to give $Cl_2(g)$, which is toxic.

$$\underbrace{Cl^-(aq) + OCl^-(aq)}_{\text{from household bleach}} + 2H^+(aq) \rightleftharpoons Cl_2(g) + H_2O(l)$$

Reaction with sodium hydroxide solution

When we pass chlorine into cold and dilute sodium hydroxide solution, sodium chloride (NaCl) and sodium hypochlorite (NaOCl) are formed. N14 N15



The oxidation number of chlorine element is 0. Oxidation numbers of Cl in NaCl and NaOCl are –1 and +1 respectively. Thus, chlorine is simultaneously reduced and oxidized.

On the other hand, chlorine reacts with hot and concentrated sodium hydroxide solution to give sodium chloride (NaCl) and sodium chlorate ($NaClO_3$).

$$\overset{0}{3Cl_2(g)} + 6NaOH(aq) \longrightarrow \overset{-1}{5NaCl(aq)} + \overset{+5}{NaClO_3(aq)} + 3H_2O(l)$$

The oxidation number of chlorine element is 0. Oxidation numbers of Cl in NaCl and $NaClO_3$ are –1 and +5 respectively. Thus, chlorine is simultaneously reduced and oxidized.

A reaction in which the same species is simultaneously reduced and oxidized is called **disproportionation**. N16

N14 Examination questions often ask about the reaction between chlorine and sodium hydroxide solution.

N15 **Chlorine bleach**
Chlorine bleach is made by reacting chlorine with sodium hydroxide solution. The active ingredient is sodium hypochlorite.
When chlorine bleach mixes with a dilute acid, chlorine gas will be released.

$$\underbrace{Cl^-(aq) + OCl^-(aq)}_{\text{from chlorine bleach}} + 2H^+(aq) \longrightarrow Cl_2(g) + H_2O(l)$$

N16 Refer to Teacher's Resources Book page 23.

Practice 20.6

P20.6 Refer to Teacher's Resources Book page 42.

Suggest a chemical test to distinguish between solutions of potassium chloride and potassium iodide. State your expected observations.

disproportionation 歧化作用

Video
Comparing the action of nitric acid of different concentrations on magnesium and copper

20.3
Investigating the action of nitric acid of different concentrations on metals.

20.16 Nitric acid of different concentrations as oxidizing agents

In Topic 4 Acids and Bases, we have learnt that only very dilute nitric acid acts like a typical acid. Both dilute and concentrated **nitric acids** are strong oxidizing agents. They can oxidize most metals, even those below hydrogen in the electrochemical series (e.g. copper).



Fig. 20.15 Brown nitrogen dioxide gas is produced when concentrated nitric acid reacts with copper

Caution: Carry out the experiment inside a fume cupboard as a toxic gas (NO_2) is given off.



Fig. 20.16 Concentrated nitric acid is a strong oxidizing agent

Caution: Carry out the experiments inside a fume cupboard as toxic gases are given off.

A further reaction may occur:

$$SO_2(g) + 2HNO_3(aq) \longrightarrow H_2SO_4(aq) + 2NO_2(g)$$

Concentrated nitric acid as an oxidizing agent

N17 Refer to Teacher's Resources Book page 24.

Action of concentrated nitric acid on metals

CE 2005 Paper 2 Q33 CE 2007 Paper 2 Q46

Consider the action of concentrated nitric acid (about 15 mol dm^{-3}) on copper (Fig. 20.15). The nitrate ions are reduced to nitrogen dioxide (NO_2). The oxidation number of nitrogen decreases from +5 to +4.

The ionic half-equation for the reduction of nitrate ions is:

$$\overset{+5}{NO_3^-}(aq) + 2H^+(aq) + e^- \longrightarrow \overset{+4}{NO_2}(g) + H_2O(l)$$

The balanced redox equation for the reaction is:

$$Cu(s) + 2NO_3^-(aq) + 4H^+(aq) \longrightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$$

The corrosive nature of concentrated nitric acid is due to its powerful oxidizing property (Fig. 20.16). Brown nitrogen dioxide gas is very toxic and corrosive.

Action of concentrated nitric acid on non-metals

Hot concentrated nitric acid can oxidize non-metals such as carbon and sulphur.

$$C(s) + 4HNO_3(aq) \longrightarrow CO_2(g) + 4NO_2(g) + 2H_2O(l)$$

$$S(s) + 4HNO_3(aq) \longrightarrow SO_2(g) + 4NO_2(g) + 2H_2O(l)$$

AL 2006 Paper 1 Q3(b)(i)

nitric acid 硝酸

Caution: Carry out the experiment inside a fume cupboard as toxic gases (NO and NO_2) are given off.

N18 Refer to Teacher's Resources Book page 24.

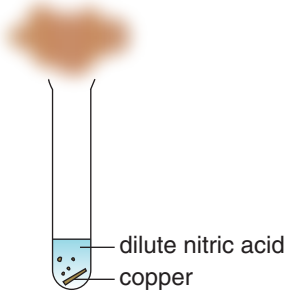


Fig. 20.17 Colourless nitrogen monoxide is formed when dilute nitric acid reacts with copper; this gas reacts with oxygen in the air to give brown nitrogen dioxide gas

N19 Dilute nitric acid can convert iron(II) ions to iron(III) ions.

Dilute nitric acid as an oxidizing agent N18

CE 2011 Paper 2 Q49

Consider the reaction between dilute nitric acid (about 1.5 mol dm^{-3}) and copper. The nitrate ions are reduced to nitrogen monoxide (NO). The oxidation number of nitrogen decreases from +5 to +2. N19

CE 2006 Paper 2 Q38

The ionic half-equation for the reduction of nitrate ions is:

AL 2005 Paper 2 Q4(c)

$$\overset{+5}{NO_3^-}(aq) + 4H^+(aq) + 3e^- \longrightarrow \overset{+2}{NO}(g) + 2H_2O(l)$$

The balanced redox equation for the reaction is:

$$3Cu(s) + 2NO_3^-(aq) + 8H^+(aq) \longrightarrow 3Cu^{2+}(aq) + 2NO(g) + 4H_2O(l)$$

CE 2006 Paper 2 Q38

Nitrogen monoxide gas is colourless. When this gas mixes with air, it reacts with oxygen in the air to give brown nitrogen dioxide gas. We can observe a brown gas at the opening of the test tube (Fig. 20.17).

$$\underset{\text{colourless}}{2NO(g)} + O_2(g) \longrightarrow \underset{\text{brown}}{2NO_2(g)}$$

Practice 20.7 P20.7 Refer to Teacher's Resources Book page 42.

Complete the following table to compare the action of nitric acid of different concentrations on metals.

Acid	Action of acid on magnesium	Action of acid on copper	Property shown by the acid (acidic / oxidizing property)
	Name of gas given off, if any	Name of gas given off, if any	
Very dilute nitric acid			
Dilute nitric acid			
Concentrated nitric acid			

Video
Action of concentrated sulphuric acid on zinc and copper

20.4
Investigating the properties of concentrated sulphuric acid.

N20 Refer to Teacher's Resources Book page 25.

We represent concentrated sulphuric acid by $H_2SO_4(l)$. This is because it contains about 98% sulphuric acid by mass.

20.17 Concentrated sulphuric acid as an oxidizing agent

Dilute **sulphuric acid** shows all the typical properties of a dilute acid. It reacts with zinc to give hydrogen and shows no reaction with copper. N20

Concentrated sulphuric acid◂ is a powerful oxidizing agent. It is reduced to sulphur dioxide in the reaction. The oxidation number of sulphur decreases from +6 to +4. N21

$$2\overset{+6}{H_2SO_4}(l) + 2e^- \longrightarrow SO_4^{2-}(aq) + \overset{+4}{SO_2}(g) + 2H_2O(l)$$

N21 Sulphur dioxide can be absorbed by sodium hydroxide solution.
$SO_2(g) + 2NaOH(aq) \longrightarrow Na_2SO_3(aq) + H_2O(l)$

Action of concentrated sulphuric acid on metals

CE 2005 Paper 2 Q33 CE 2009 Paper 1 Q6(b)(i)

Hot concentrated sulphuric acid can oxidize most metals. For example, it reacts with copper to give sulphur dioxide gas◂ (Fig. 20.18). N22

$$Cu(s) + 2H_2SO_4(l) \longrightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$$

N22 Refer to Teacher's Resources Book page 26.

Caution: Carry out the experiment inside a fume cupboard as a toxic gas (SO_2) is given off.



Fig. 20.18 Action of concentrated sulphuric acid on copper

Action of concentrated sulphuric acid on non-metals

Hot concentrated sulphuric acid can oxidize non-metals such as carbon and sulphur◂.

$$C(s) + 2H_2SO_4(l) \longrightarrow CO_2(g) + 2SO_2(g) + 2H_2O(l)$$

$$S(s) + 2H_2SO_4(l) \longrightarrow 3SO_2(g) + 2H_2O(l)$$

CE 2007 Paper 2 Q20, Q21

Caution: Carry out the experiments inside a fume cupboard as a toxic gas (SO_2) is given off.

N23 Refer to Teacher's Resources Book page 26.

Video
Action of concentrated sulphuric acid on halides

Action of concentrated sulphuric acid on halides N23

Sodium chloride

Concentrated sulphuric acid liberates steamy fumes of hydrogen chloride when it reacts with sodium chloride.

$$NaCl(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HCl(g)$$◂

AL 2006 Paper 2 Q1(b)(iii)

This is NOT a redox reaction because the oxidation numbers of all the elements remain unchanged in the reaction.

sulphuric acid 硫酸

Sodium bromide

When concentrated sulphuric acid is added to sodium bromide, hydrogen bromide is given off.

$$NaBr(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HBr(g)$$

Hydrogen bromide is a stronger reducing agent than hydrogen chloride. It reduces the concentrated sulphuric acid to sulphur dioxide. Hydrogen bromide is oxidized to bromine (Fig. 20.19).

$$\overset{-1}{2HBr}(g) + \overset{+6}{H_2SO_4}(l) \longrightarrow \overset{0}{Br_2}(g) + \overset{+4}{SO_2}(g) + 2H_2O(l)$$

reddish brown fumes

Caution: Carry out the experiment inside a fume cupboard as toxic gases (Br_2 and SO_2) are given off.



Fig. 20.19 Bromine fumes are produced when concentrated sulphuric acid reacts with sodium bromide

Sodium iodide

When concentrated sulphuric acid is added to sodium iodide, several reactions occur. In one of the reactions, hydrogen iodide is produced.

$$NaI(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HI(g)$$

N24 HI(g) cannot be prepared by treating KI(s) with concentrated H_2SO_4 as the HI(g) formed will be oxidized by concentrated H_2SO_4 to I_2(s).
Appropriate method: treat KI(s) with concentrated H_3PO_4.
$KI(s) + H_3PO_4(l) \longrightarrow KH_2PO_4(s) + HI(g)$

Hydrogen iodide is a stronger reducing agent than hydrogen chloride and hydrogen bromide. In the predominant reaction, hydrogen iodide reduces the concentrated sulphuric acid further, producing hydrogen sulphide gas. Hydrogen iodide is oxidized to iodine (Fig. 20.20). N24

$$\overset{-1}{8HI}(g) + \overset{+6}{H_2SO_4}(l) \longrightarrow \overset{0}{4I_2}(s) + \overset{-2}{H_2S}(g) + 4H_2O(l)$$

black

The purple iodine fumes condense to a black solid.

Caution: Hydrogen sulphide is a toxic gas with a bad egg smell. Carry out the experiment inside a fume cupboard.

ASL 2006 Q3(b)(i)
AL 2006 Paper 2 Q1(b)(iii)

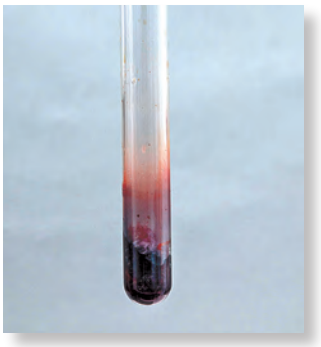


Fig. 20.20 Iodine fumes are produced when concentrated sulphuric acid reacts with sodium iodide

Practice 20.8

P20.8 Refer to Teacher's Resources Book page 42.

Complete the following table to compare the properties of dilute and concentrated sulphuric acid.

Action of acid on	Dilute sulphuric acid		Concentrated sulphuric acid	
	Name of gas given off, if any	Property shown by the acid (acidic / oxidizing property)	Name of gas given off, if any	Property shown by the acid (acidic / oxidizing property)
Sodium carbonate				
Zinc				
Copper				

Video
- Investigating the chemical properties of sulphur dioxide (I) (microscale experiment)
- Investigating the chemical properties of sulphur dioxide (II) (microscale experiment)

N25 Examination questions often ask the reducing properties of sulphite solution / aqueous sulphur dioxide, and colour changes of the ions involved.

20.18 Aqueous sulphur dioxide as a reducing agent

Aqueous **sulphur dioxide** is a good reducing agent. Sulphur dioxide N25 dissolves in water to form sulphurous acid.

$$SO_2(g) + H_2O(l) \rightleftharpoons \underset{\text{sulphurous acid}}{H_2SO_3(aq)}$$

The reducing properties of aqueous sulphur dioxide are due to the presence of sulphite ions (SO_3^{2-}) from sulphurous acid. In a redox reaction, the sulphite ions are oxidized to sulphate ions (SO_4^{2-}). The oxidation number of sulphur increases from +4 to +6.

CE 2009 Paper 2 Q49

$$\overset{+4}{SO_3^{2-}}(aq) + H_2O(l) \longrightarrow \overset{+6}{SO_4^{2-}}(aq) + 2H^+(aq) + 2e^-$$

sulphur dioxide 二氧化硫

Table 20.10 summarizes reactions which occur when we bubble sulphur dioxide into solutions of some oxidizing agents. N26

CE 2007 Paper 2 Q22 CE 2008 Paper 2 Q21

Table 20.10

Reducing actions of aqueous sulphur dioxide

Reaction of SO_2 / SO_3^{2-} with	Observation	Equation
$MnO_4^-(aq)/H^+(aq)$	purple → pale pink / colourless	DSE 2014 Paper 1B Q9(b) $5SO_3^{2-}(aq) + 2MnO_4^-(aq) + 6H^+(aq) \longrightarrow 5SO_4^{2-}(aq) + 2Mn^{2+}(aq) + 3H_2O(l)$
$Cr_2O_7^{2-}(aq)/H^+(aq)$	orange → green	$3SO_3^{2-}(aq) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \longrightarrow 3SO_4^{2-}(aq) + 2Cr^{3+}(aq) + 4H_2O(l)$
$Br_2(aq)$	yellow-brown → colourless	$SO_3^{2-}(aq) + H_2O(l) + Br_2(aq) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2Br^-(aq)$
$Fe^{3+}(aq)$	yellow-brown → pale green	$SO_3^{2-}(aq) + H_2O(l) + 2Fe^{3+}(aq) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2Fe^{2+}(aq)$

CE 2009 Paper 2 Q24 CE 2011 Paper 2 Q26 DSE 2012 Paper 1A Q22 CE 2010 Paper 2 Q48 CE 2011 Paper 2 Q26 N27



Fig. 20.21 A piece of filter paper soaked with acidified potassium dichromate solution turns from orange to green in the presence of sulphur dioxide

We can use a piece of filter paper soaked with **acidified potassium dichromate solution** to test for sulphur dioxide gas. The paper changes from orange to green if sulphur dioxide is present (Fig. 20.21). Sulphur dioxide reduces the orange dichromate ions to green chromium(III) ions. N28

CE 2007 Paper 2 Q20
AL 2005 Paper 1 Q7(b)(iii)

$$\underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + 3SO_2(aq) + 2H^+(aq) \longrightarrow \underset{\text{green}}{2Cr^{3+}(aq)} + 3SO_4^{2-}(aq) + H_2O(l)$$

N26 Uses of sulphur dioxide

- Making sulphuric acid
  The most important use of sulphur dioxide is as an intermediate in the production of sulphuric acid.
- As an industrial bleach
  Sulphur dioxide is a common industrial bleach. It is used to bleach delicate materials such as wool, silk, paper, which would otherwise be damaged by chlorine. It is also used to bleach wood pulp for making newspapers.
- As a preservative
  Sulphur dioxide is added into a wide range of foods such as fruit juice and dried fruits as a preservative.
  In the manufacture of wine, a small amount of sulphur dioxide is added to control the growth of unwanted microbes.

N27 Refer to Teacher's Resources Book page 27.

N28 Refer to Teacher's Resources Book page 27.

acidified potassium dichromate solution 酸化重鉻酸鉀溶液

Example 20.9

Q Sulphur dioxide can be obtained from the reaction between hot concentrated sulphuric acid and copper turnings.

a) Draw a labelled diagram to show the set-up in preparing and collecting sulphur dioxide.

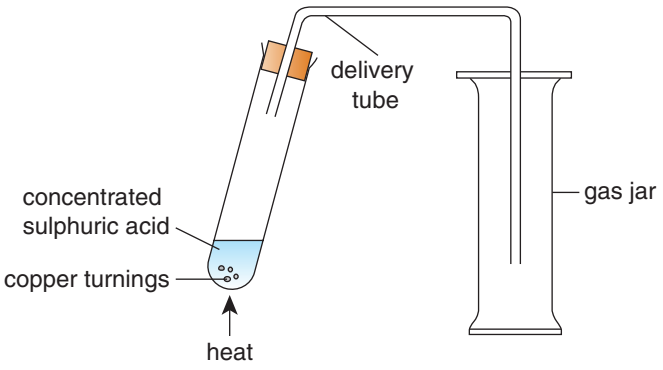
b) State the observation and write a chemical equation for the reaction that occurs.

CE 2010 Paper 1 Q7(c)

c) Some people would use sulphur dioxide to treat food.

Suggest a purpose of treating food with sulphur dioxide.

A a)



b) Any one of the following:

- Copper dissolves.
- A blue solution results.
- A colourless gas evolves.
- A choking smell is detected.

$Cu(s) + 2H_2SO_4(l) \longrightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$

c) Food preservation

Practice 20.9

P20.9 Refer to Teacher's Resources Book page 43.

Suggest a chemical test to distinguish between solutions of sodium sulphite and sodium sulphate. State your expected observation.

DSE 2013 Paper 1A Q22

Have you mastered?

Key terms

oxidation 氧化作用 36 reduction 還原作用 36 redox reaction 氧化還原反應 37

reducing agent 還原劑 39 oxidizing agent 氧化劑 39 oxidation number 氧化數 41

acidified potassium permanganate solution 酸化高錳酸鉀溶液 50

acidified potassium dichromate solution 酸化重鉻酸鉀溶液 50 breathalyzer 呼氣分析儀 51

redox equation 氧化還原方程式 52 disproportionation 歧化作用 66

Checklist

After studying this unit, you should be able to

- ☐ identify redox reactions in terms of loss and gain of electron(s);
- ☐ identify redox reactions in terms of changes in oxidation numbers;
- ☐ identify the oxidizing agent and the reducing agent in a redox reaction;
- ☐ assign oxidation numbers to elements in chemical species;
- ☐ write balance redox equations;
- ☐ describe the oxidizing actions of aqueous chlorine, nitric acids of different concentrations and concentrated sulphuric acid;
- ☐ describe the reducing action of aqueous sulphur dioxide.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The following table summarizes the definitions of oxidation and reduction.

Defining in terms of	Oxidation	Reduction
Gaining and losing oxygen	process in which a species gains oxygen	process in which a species loses oxygen
Gaining and losing hydrogen	process in which a species loses hydrogen	process in which a species gains hydrogen
Electron transfer	process in which a species loses electrons	process in which a species gains electrons
Change in oxidation numbers	process in which the oxidation number of an element in a species increases	process in which the oxidation number of an element in a species decreases

2 Oxidation and reduction always takes place together. The combined process is called a redox reaction.

3 The following table summarizes the characteristics of reducing and oxidizing agents.

Reducing agent	Oxidizing agent
causes reduction	causes oxidation
loses electrons	gains electrons
causes a decrease in the oxidation number of an element in another species	causes an increase in the oxidation number of an element in another species
is oxidized	is reduced

4 The oxidation number of an atom is an imaginary oxidation number assigned to the atom according to a set of rules.

5 Obtain balanced equations for redox reactions by:

a) combining ionic half-equations; and

b) the oxidation number method.

6 In the electrochemical series,

a) the oxidizing power of oxidizing agents increases down the series;

b) the reducing power of reducing agents decreases down the series.

Have you mastered?

7 a) The following table summarizes the oxidizing / reducing properties of chlorine, nitric acid of different concentrations and concentrated sulphuric acid.

	Property	Reaction	Equation(s)
Chlorine	oxidizing property	aqueous chlorine oxidizes bromide ions to bromine	$Cl_2(aq) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(aq)$
		aqueous chlorine oxidizes iodide ions to iodine	$Cl_2(aq) + 2I^-(aq) \longrightarrow 2Cl^-(aq) + I_2(aq)$
	being oxidized and reduced simultaneously (disproportionation)	reaction with cold and dilute sodium hydroxide solution	$Cl_2(g) + 2NaOH(aq) \longrightarrow NaCl(aq) + NaOCl(aq) + H_2O(l)$
		reaction with hot and conc. sodium hydroxide solution	$3Cl_2(g) + 6NaOH(aq) \longrightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$
Conc. / dilute nitric acid	oxidizing property	conc. nitric acid oxidizes metals	$Cu(s) + 2NO_3^-(aq) + 4H^+(aq) \longrightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$
		hot conc. nitric acid oxidizes non-metals	$C(s) + 4HNO_3(aq) \longrightarrow CO_2(g) + 4NO_2(g) + 2H_2O(l)$ $S(s) + 4HNO_3(aq) \longrightarrow SO_2(g) + 4NO_2(g) + 2H_2O(l)$
		dilute nitric acid oxidizes metals	$3Cu(s) + 2NO_3^-(aq) + 8H^+(aq) \longrightarrow 3Cu^{2+}(aq) + 2NO(g) + 4H_2O(l)$
Conc. sulphuric acid◀	oxidizing property	can oxidize most metals	$Cu(s) + 2H_2SO_4(l) \longrightarrow CuSO_4(aq) + SO_2(g) + 2H_2O(l)$
		can oxidize most non-metals	$C(s) + 2H_2SO_4(l) \longrightarrow CO_2(g) + 2SO_2(g) + 2H_2O(l)$ $S(s) + 2H_2SO_4(l) \longrightarrow 3SO_2(g) + 2H_2O(l)$
		reaction with sodium bromide	$NaBr(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HBr(g)$ $2HBr(g) + H_2SO_4(l) \longrightarrow Br_2(g) + SO_2(g) + 2H_2O(l)$
		reaction with sodium iodide	$NaI(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HI(g)$ $8HI(g) + H_2SO_4(l) \longrightarrow 4I_2(s) + H_2S(g) + 4H_2O(l)$

◀ The reaction between conc. sulphuric acid and sodium chloride is NOT a redox reaction.

Have you mastered?

b) The following table summarizes the reducing properties of aqueous sulphur dioxide.

	Property	Reaction	Equation(s)
Aqueous sulphur dioxide	reducing property	aqueous sulphur dioxide reduces dichromate ions to chromium(III) ions	$3SO_3^{2-}(aq) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \longrightarrow 3SO_4^{2-}(aq) + 2Cr^{3+}(aq) + 4H_2O(l)$

Unit Exercise

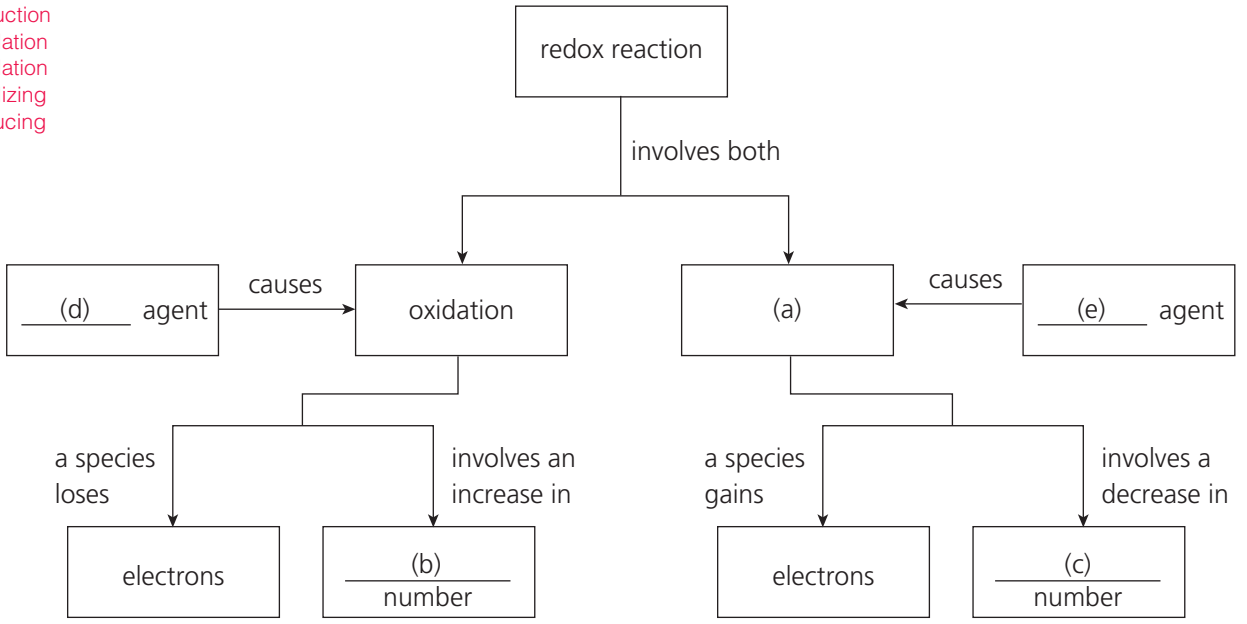
Refer to Teacher's Resources Book pages 43–49.

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

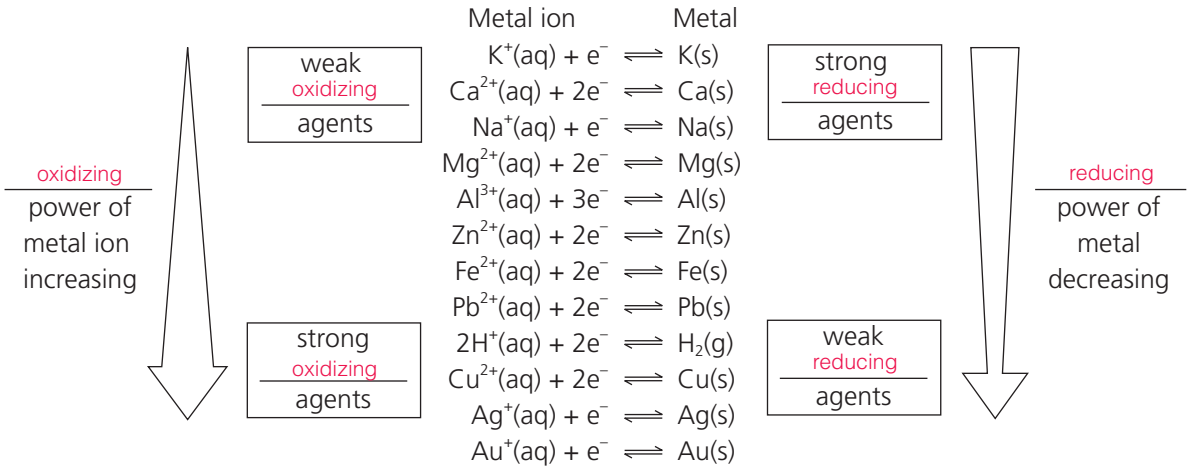
a) reduction
b) oxidation
c) oxidation
d) oxidizing
e) reducing



2 Complete the following table on the characteristics of oxidizing and reducing agents.

	Species that causes	Species that is
Reducing agent	oxidation / reduction	oxidized / reduced
Oxidizing agent	oxidation / reduction	oxidized / reduced

3 Fill in the blanks for the following electrochemical series.



4 a) Complete the following table on the reactions of some common oxidizing agents.

Oxidizing agent	Product when reduced	Ionic half-equation
Aqueous chlorine	$Cl^-(aq)$	$Cl_2(aq) + 2e^- \longrightarrow 2Cl^-(aq)$
Aqueous bromine	$Br^-(aq)$	$Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq)$
Potassium permanganate in acidic solution	$Mn^{2+}(aq)$	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$
Potassium dichromate in acidic solution	$Cr^{3+}(aq)$	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$
Iron(III) sulphate solution	$Fe^{2+}(aq)$	$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$

b) Complete the following table on the reactions of some common reducing agents.

Reducing agent	Product when oxidized	Ionic half-equation
Sodium sulphite solution	$SO_4^{2-}(aq)$	$SO_3^{2-}(aq) + H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$
Iron(II) sulphate solution	$Fe^{3+}(aq)$	$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$
Potassium bromide solution	$Br_2(aq)$	$2Br^-(aq) \longrightarrow Br_2(aq) + 2e^-$
Potassium iodide solution	$I_2(aq)$	$2I^-(aq) \longrightarrow I_2(aq) + 2e^-$

Part II Multiple choice questions

Refer to Teacher's Resources Book for explanations.

5 What is the oxidation number of Cu in $Cu(NH_3)_4Cl_2$?

A 0
(B) +2
C +4
D +6

(HKDSE, Paper 1A, 2012, 6)

6 In which of the following compounds is the oxidation number of nitrogen lowest?

(A) NH_4Cl
B $NaNO_2$
C NH_2OH
D $Pb(NO_3)_2$

(HKCEE, Paper 2, 2008, 5)

7 The following three chlorine-containing species are arranged according to the increasing order of oxidation number of chlorine:

HCl, J, $HClO_3$

Which of the following species may NOT be J?

A Cl_2
B ClO_2^-
(C) Cl_2O_7
D HOCl

(HKCEE, Paper 2, 2011, 11)

8 Consider the ionic equation below:

$$2MnO_4^- + xSn^{2+} + yH^+ \longrightarrow 2Mn^{2+} + xSn^{4+} + 8H_2O$$

What is the value of x ?

A 2
B 4
(C) 5
D 7

(HKCEE, Paper 2, 2008, 4)

9 X and Z are metals. X reacts with $Z(NO_3)_2$ solution according to the following equation:

$$X(s) + Z^{2+}(aq) \longrightarrow X^{2+}(aq) + Z(s)$$

Which of the following deductions is correct?

A Both X and Z can react with water.
B The reactivity of Z is higher than that of X.
(C) X acts as a reducing agent in the reaction.
D Z acts as the negative pole when X and Z are used as electrodes in a chemical cell with sodium chloride solution as electrolyte.

(HKCEE, Paper 2, 2008, 15)

10 Chlorine is bubbled into an aqueous solution P for some time. The colour of P gradually changes to brown. P is NOT likely to be

(A) calcium hydroxide solution.
B potassium iodide solution.
C iron(II) chloride solution.
D zinc bromide solution.

(HKCEE, Paper 2, 2007, 19)

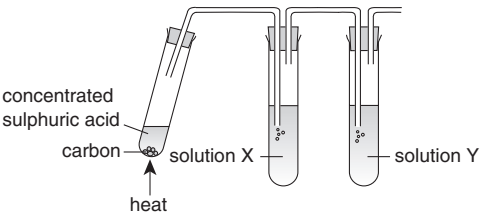
11 In which of the following processes does sulphur dioxide act as a reducing agent?

A Passing sulphur dioxide into water
(B) Passing sulphur dioxide into iodine solution
C Passing sulphur dioxide into iron(II) sulphate solution
D Passing sulphur dioxide into sodium hydroxide solution

(HKCEE, Paper 2, 2008, 11)

Directions: Questions 12 and 13 refer to the following experiment.

When carbon and concentrated sulphuric acid are heated together, sulphur dioxide and carbon dioxide are produced. These two gases are then passed through solution X and solution Y as shown below:



Unit Exercise

12 Which of the following combinations of solution X and solution Y can be used to show that sulphur dioxide and carbon dioxide are produced?

	Solution X	Solution Y
(A)	bromine water	calcium hydroxide solution
B	iron(II) sulphate solution	calcium hydroxide solution
C	acidified potassium dichromate solution	sodium hydroxide solution
D	acidified potassium permanganate solution	sodium hydroxide solution

(HKCEE, Paper 2, 2007, 20)

13 Which of the following sttatements concerning the reaction between carbon and concerntrated sulphuric acid are correct?

(1) The oxidation number of carbon changes from 0 to +4.
(2) The oxidation number of hydrogen in sulphuric acid remains unchanged.
(3) Concentrated sulphuric acid acts both as a dehydrating agent and an oxidizing agent.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2007, 21)

14 Which of the following is /are redox reaction(s)?

(1) $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$
(2) $2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$
(3) $2KNO_3 \longrightarrow 2KNO_2 + O_2$

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2010, 24)

15 Consider the redox reaction represented by the equation below:

$$PbO_2 + NO_2^- + 2H^+ \longrightarrow Pb^{2+} + NO_3^- + H_2O$$

Which of the following statements is / are correct?

(1) NO_2^- is reduced.
(2) PbO_2 is reduced.
(3) H^+ is neither oxidized nor reduced.

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKCEE, Paper 2, 2011, 21)

16 Which of the following reagents can convert iron(II) ions to iron(III) ions?

(1) Chlorine water
(2) Dilute nitric acid
(3) Potassium bromide solution

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2010, 47)

17 Which of the following reagents can be used to distinguish between sodium sulphite and sodium sulphate?

(1) Iron(II) chloride solution
(2) Acidified potassium permanganate solution
(3) Concentrated nitric acid

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Paper 1A, 2013, 22)

Part III Structured questions

18 Find the oxidation numbers of the underlined elements in the following species:

a) $\underline{S}O_3$

b) $\underline{Mn}O_4^-$

c) $H\underline{C}O_3^-$

d) $H_2\underline{S}_2O_7$

e) $\underline{Pb}SO_4$

f) $Ca\underline{H}_2$

g) $[\underline{Al}(OH)_4]^-$

h) $H_2\underline{O}_2$

i) $[\underline{Co}Cl_4]^{2-}$

j) $\underline{V}O_2^+$

19 Which of the following equations represent redox reactions? Specify the species being reduced in each case if applicable.

a) $NaI(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HI(g)$

b) $2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$

c) $2NH_3(g) + 3CuO(s) \longrightarrow 3Cu(s) + N_2(g) + 3H_2O(l)$

d) $Zn(s) + 2AgNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + 2Ag(s)$

e) $2KBr(aq) + Cl_2(aq) \longrightarrow 2KCl(aq) + Br_2(aq)$

20 Balance the following equations for reactions that occur in acidic conditions:

a) $Cr_2O_7^{2-}(aq) + I^-(aq) \longrightarrow Cr^{3+}(aq) + I_2(aq)$

b) $MnO_4^-(aq) + Cl^-(aq) \longrightarrow Mn^{2+}(aq) + Cl_2(g)$

c) $Fe^{2+}(aq) + H_2O_2(aq) \longrightarrow Fe^{3+}(aq) + H_2O(l)$

21 For each of the reactions below,

i) state and explain what would be observed;
ii) write an appropriate equation.

a) Adding excess sodium sulphite solution to iodine solution

b) Adding aqueous chlorine to potassium bromide solution

c) Adding excess potassium iodide solution to acidified potassium permanganate solution

d) Adding excess iron(II) sulphate solution to acidified potassium dichromate solution

e) Adding concentrated nitric acid to magnesium ribbons

f) Adding dilute nitric acid to zinc granules

g) Adding concentrated sulphuric acid to solid sodium bromide

22 In the Periodic Table, the chemistry of elements in a group can often be predicted from the chemistry of just one element in the group.

Ions of Group 7 elements take part in displacement reactions. These reactions can be used to compare the reactivities of the elements within Group 7.

A student adds aqueous solutions of halogens to test tubes containing solutions of halide ions. The resulting mixtures are then shaken with cyclohexane, an organic solvent.

One of the student's results is shown in the table.

Experiment number	Experiment details	Colour seen within the organic solvent
1	addition of $Cl_2(aq)$ to $I^-(aq)$ ions	
2	addition of $Cl_2(aq)$ to $Br^-(aq)$ ions	orange
3	addition of $Br_2(aq)$ to $Cl^-(aq)$ ions	

a) Complete the table to show the expected colours.

b) Write the ionic equation for the reaction taking place in *Experiment 2*.

c) These three experiments alone are unable to confirm the order of reactivity for Cl_2, Br_2 and I_2.

Suggest ONE further displacement reaction which could be carried out to confirm the order of reactivity of Cl_2, Br_2 and I_2.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2010, 4(a))

Unit Exercise

23 a) Explain the term reducing agent in terms of oxidation number change.

b) Write ionic half-equations to show:

i) hypochlorite ions, OCl^-, in acidic solution, being reduced to chlorine molecules and water;

ii) chloride ions being oxidized to chlorine molecules.

c) Combine the two equations in (b) to show the effect of adding an acid to a mixture of hypochlorite ions and chloride ions.

d) Describe what you would see if concentrated sulphuric acid is added to solid sodium iodide.

e) Potassium chlorate, $KClO_3$, decomposes on heating to give potassium chloride, KCl, and oxygen, O_2.

i) Write the equation for this reaction.

ii) Show, by the use of oxidation numbers, why this is a redox reaction.

(Edexcel Advanced Subsidiary GCE, Unit Test 1, Jan. 2008, 4)

24 Traditional gunpowder is a mixture of potassium nitrate, sulphur powder and charcoal powder. When this gunpowder is ignited, the reaction that occurs can be represented by the following equation:

$$2KNO_3(s) + S(s) + 3C(s) \longrightarrow K_2S(s) + N_2(g) + 3CO_2(g)$$

a) In terms of oxidation number, explain whether the reaction involves oxidation and reduction.

b) Based on the above information, suggest why traditional gunpowder is explosive.

c) Draw the electronic diagram of K_2S, showing electrons in the *outermost* shells only.

(HKCEE, Paper 1, 2010, 8(a), (c), (d))

25 a) Sulphur dioxide reacts with sodium carbonate solution to form sodium hydrogensulphite ($NaHSO_3$). $NaHSO_3$ is commonly added to red wine for preventing the ethanol in the wine from turning to ethanoic acid.

i) State the oxidation number of sulphur in $NaHSO_3$.

ii) In terms of oxidation and reduction, explain how $NaHSO_3$ can prevent ethanol from turning to ethanoic acid.

iii) 0.1 mole of sulphur dioxide is dissolved in excess sodium carbonate solution to form $NaHSO_3$ solution. Calculate the mass of $NaHSO_3$ formed. (Assume that sulphur dioxide is completely converted to $NaHSO_3$.)

(Relative atomic masses: H = 1.0, O = 16.0, Na = 23.0, S = 32.1)

b) Sodium hydrogensulphite ($NaHSO_3$) reacts with zinc to form sodium hydrosulphite ($Na_2S_2O_4$) and zinc hydroxide only. $Na_2S_2O_4$ is commonly used to bleach paper.

i) Write a chemical equation for the reaction of $NaHSO_3$ with zinc.

ii) What is the role of zinc in the reaction?

(HKCEE, Paper 1, 2011, 5(a)–(b))

26 For each of the following pairs of substances, suggest a chemical test to distinguish one substance from the other and state the expected observations.

a) Sodium chloride and sodium iodide

b) Dilute sulphuric acid and dilute nitric acid

c) Iron(II) sulphate solution and iron(III) sulphate solution

27 a) State the observation and write a chemical equation for the reaction between hot concentrated sulphuric acid and copper turnings.

b) Hot concentrated sulphuric acid reacts with copper turnings inside a test tube. Describe how you should clean the test tube after the reaction.

(HKCEE, Paper 1, 2009, 6(b))